Temporal Dependence and Stationary Bias in Adaptive Optimization

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Abstract

Adaptive optimizers like RMSprop and Adam divide each step by a memory of recent squared gradients, so the order in which data arrive can change where they settle. We compare a stationary finite-state Markov source with independent sampling from the same marginal law. The objective and the distribution of each sample stay fixed, and only the temporal dependence changes. In practice the momentum and variance memories β_1 and β_2 stay fixed while the learning rate is tuned. The difference between the two sources then does not have to vanish with the learning rate — dependence moves the averaged equilibrium itself. With β_2 close to one, a momentum-weighted sum of lag correlations between gradients and squared gradients gives this displacement to leading order. The same sum decides whether dependence lowers or raises the limiting loss, and both happen even for reversible sources. These results are for scalar quadratic losses with a smoothed denominator, and the displaced equilibrium extends to strongly convex losses in any dimension when the gradient noise is additive. Comparing a source with its time reversal, which keeps the long-run covariance, isolates the part of the displacement that depends on the direction of time. A source can match independent sampling in both its marginal law and its long-run covariance and still move where the optimizer settles.

Keywords. Adaptive optimization; Markovian stochastic approximation; stationary bias; temporal dependence; time reversal; stochastic averaging.

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1 Introduction

Adaptive optimizers carry a record of earlier gradients into each update. RMSprop and Adam divide every gradient by a running memory of squared gradients, so a step that arrives right after a run of large gradients is damped, and one that arrives after small gradients is not. Streaming observations, trajectories collected under a fixed policy, and repeated uses of the same sample can all produce temporally dependent data. The order of the samples can then change where the optimizer ends up, even if the objective and the distribution of single samples stay fixed. Convergence theory for dependent sampling asks how much dependence slows an optimizer down [14, 16]. We ask instead how it moves the point an adaptive optimizer settles around.

We isolate this effect with one comparison — the same optimizer, run on a stationary finite-state Markov source P and on independent sampling Π from its stationary law. The difference between the two stationary parameter means is the *dependence shift*. In RMSprop and Adam it comes from lag correlations between gradients and squared gradients, and with fixed memories it survives as a displaced equilibrium when the learning rate goes to zero.

A two-state example shows how this works. Let the gradient noise be 1 with probability $2/3$ and -2 with probability $1/3$. Under independent sampling, the rare large gradient pushes the parameter up, but it also inflates the variance memory and so damps its own step. The frequent small gradients push the parameter down and are damped less, so RMSprop settles below the minimizer. Now let the source repeat its last draw with probability ρ and draw a fresh sample otherwise. The marginal law stays the same, but gradients arrive in runs, and each large gradient also damps its repeats. With fixed memories and β_2 close to one, the RMSprop displacement grows by the factor $1/(1-\rho)$ — the expected number of times each draw is used. A source that switches more often than independent sampling shrinks the displacement instead. Both sources are reversible, so dependence can lower or raise the limiting loss with no preferred direction of time (Section 4.1).

Main results

What matters for RMSprop and Adam is how squared gradients line up in time with the gradients around them. Our results use the scalar losses $\ell(\theta, x) = \frac{1}{2}(\theta + c(x))^2$, where c is the gradient noise. Write $c_k = c(X_k)$ and $q_k = c_k^2 - \sigma^2$, where $\mathbb{E}[c_k] = 0$ and $\sigma^2 = \mathbb{E}[c_k^2]$. Every source then has the same objective and the same minimizer, $\theta = 0$. Plain SGD on these losses has stationary mean zero under every source, so any shift comes from the adaptive denominator. The lag correlations between c and q enter through the momentum-weighted lag sum

$$\mathcal{T}_Q = \mathbb{E}[c_0^3] + \sum_{s \geq 1} \big\{ \mathbb{E}_Q[q_0 c_s] + \beta_1^s \mathbb{E}_Q[c_0 q_s] \big\}.$$

Here β_1 is Adam's momentum. A squared gradient that comes s steps before a gradient counts fully. One that comes s steps after it only counts with weight β_1^s, through the part of the gradient that momentum still carries. Independent sampling has no lag terms, so $\mathcal{T}_{\Pi} = \mathbb{E}[c_0^3]$.

Our central result is for Adam with fixed memories β_1 and $\beta_2 = 1 - \alpha$, the smoothed denominator $\sqrt{v + \delta}$ with $\delta > 0$, and no weight decay or bias correction. Let μ_h^Q be an invariant law of Adam at learning rate h, driven by a source Q, in the class of Theorem 3.3. Then

$$\mathbb{E}_{\mu_h^Q}[\theta] = \theta_{Q,\alpha} + O_\alpha(h), \qquad \theta_{Q,\alpha} = \frac{\alpha}{2(\sigma^2 + \delta)}\mathcal{T}_Q + O(\alpha^2),$$

where $\theta_{Q,\alpha}$ is the stable zero of Adam's averaged drift, which we call the frozen root (Theorems 3.3 and 3.4). The first estimate holds at each fixed α as $h \to 0$, and the second as $\alpha \to 0$ afterwards. The excess loss at the frozen root is $\theta_{Q,\alpha}^2/2$, so for small α dependence lowers the limiting loss when $|\mathcal{T}_P| < |\mathcal{T}_{\Pi}|$ and raises it when the inequality goes the other way (Corollary 3.5).

The displacement does not fade with the learning rate, and that sets it apart from the usual stationary bias of constant-step stochastic approximation, which is of the order of the step [1, 17]. In that setting the whole state slows down with the step. Here the two memories turn over at fixed rates while the parameter slows down, so the parameter only sees the memories through their stationary law at each parameter value — and that law depends on the order of the data.

Time reversal is a natural control. It keeps the source's long-run covariance but swaps which of two signals tends to come first, so comparing P with its reversal $P^\dagger$ isolates the orientation component of the displacement, the part that flips sign under reversal. In the lag sum this part is a momentum-weighted signed area between the gradient and its square (Section 3.2). The cycle in Section 4.2 even matches independent sampling in its long-run covariance, and it still moves the equilibrium.

Two further results concern scope. The averaging step needs the averaged drift to have a stable zero, and we show that this holds for every two-state source at all values of β_1 and β_2, and for any finite source once δ is large enough (Proposition 3.2). The displaced equilibrium is also not tied to scalar quadratics. For a strongly convex loss in any dimension with additive gradient noise, the averaged drift of Adam vanishes exactly where the population gradient equals the vector of scalar frozen roots (Proposition 3.6).

Related work

Optimization with dependent samples has a substantial convergence theory. Duchi et al. [14] analyze mirror descent with ergodic data, and Sun, Sun, and Yin [29] study gradient descent along Markov chains. Nagaraj et al. [27] establish statistical limits and algorithms for

regression with Markovian data, and Even [16] and Beznosikov et al. [3] give convergence and complexity bounds in terms of the sampling chain.

Dieuleveut, Durmus, and Bach [13] treat the iterates of constant-step SGD with i.i.d. gradient noise as a Markov chain. Huo, Chen, and Xie [18] expand the stationary bias of linear stochastic approximation with Markovian data and use extrapolation to reduce it. Huo et al. [17] and Allmeier and Gast [1] develop bias expansions with Markovian input. Their bias is of the order of the step size. With fixed memories the leading effect is a displaced equilibrium that does not vanish with the step, and we make its lag weights explicit.

For nonlinear two-timescale recursions with Markovian noise, Lamouri, Baudry, and Gast [23] give bias bounds. Madhusudanarao and Singh [25] derive stationary-bias expansions with error bounds uniform as the slow-to-fast step ratio vanishes. Their expansion includes a term proportional to the fast step, which need not disappear when only the slow step is reduced. We keep the memories fixed, prove confinement for the scalar recursion, and compute how the source changes the averaged equilibrium.

Adam combines first- and second-moment memories [21]. Reddi, Kale, and Kumar [28] show that it can fail to converge, Zhou et al. [31] study the correlation between gradients and preconditioners, and Cattaneo, Klusowski, and Shigida [5] derive modified equations for its implicit bias. Massé and Ollivier [26] prove local convergence of RMSprop and of Adam with $\beta_2 \to 1$. Dereich and Jentzen [7] bound RMSprop's loss up to a stopping time under i.i.d. input, with explicit control of the memory parameters. Averaged Adam fields and their displaced zeros appear in Dereich and Jentzen [6], and Dereich, Jentzen, and Kassing [8] develop the matching ODE approximation with fixed memories and i.i.d. data. Dereich et al. [9] study a class of simple quadratic problems with two-point i.i.d. data and show that Adam converges to the minimizer if and only if the data are symmetrically distributed. A Markov source can have a symmetric one-time law and an asymmetric path law, which is exactly what the symmetric cycle of Section 4.2 uses.

Signed area and its nonlinear effects have a longer history. Deuschel, Orenshtein, and Perkowski [10] obtain an area correction for Markov additive functionals, Kelly [20] derives diffusion limits of stochastic recursions with rough paths, Engel, Friz, and Orenshtein [15] give Green–Kubo formulas for covariance and area, and Diamantakis, Holm, and Pavliotis [12] derive a shifted Gibbs measure in a dissipative Lie–Poisson model. Carrondo's seminar announcement [4] describes joint work with Pelizzari on rough invariance principles and random-reshuffling SGD.

Kar [19] analyzes displaced RMSprop fixed points under periodic rare-token arrivals, Sweeney [30] studies optimizer-memory order effects over finite shuffle windows, and Ziyin et al. [32] study the irreversibility of the training update itself. We instead hold the update and the distribution of each sample fixed and only change the temporal dependence of the data.

Organization. Section 2 sets up the comparison, the losses, and the lag correlations. Section 3 treats Adam with fixed memories: the averaged equilibrium, its expansion, the limiting loss, and the extension to additive noise. Section 4 collects the examples and numerics. Proofs are in the appendices.

2 Sources, Losses, and Lag Correlations

We compare three kernels with the same stationary law, and we record the lag statistics of the gradient noise that the results use.

2.1 Sources, independent sampling, and reversal

Let P be an irreducible, aperiodic transition matrix on a finite set S with stationary law π. We compare it with two kernels that have the same stationary law: independent sampling from π,

$$\Pi(x, y) = \pi(y), \tag{2.1}$$

and the stationary reversal

$$P^\dagger = \operatorname{diag}(\pi)^{-1} P^\top \operatorname{diag}(\pi). \tag{2.2}$$

Both are again irreducible and aperiodic. The source is external, so it does not react to the optimizer. We write $\mathbb{E}_Q$ for expectations under the stationary two-sided chain with kernel $Q \in \{P, \Pi, P^\dagger\}$.

Let Y be a centered vector observable on S. Its lag covariances, long-run covariance, and signed lag area are

$$C_s^Y(Q) = \mathbb{E}_Q[Y(X_0)Y(X_s)^\top], \qquad s \geq 0, \tag{2.3}$$

$$\Sigma_Y(Q) = C_0^Y(Q) + \sum_{s \geq 1} \big(C_s^Y(Q) + C_s^Y(Q)^\top\big), \tag{2.4}$$

$$\Gamma_Y(Q) = \frac{1}{2} \sum_{s \geq 1} \big(C_s^Y(Q) - C_s^Y(Q)^\top\big). \tag{2.5}$$

The earlier time always comes first, and geometric mixing makes both series converge absolutely. The area Γ_Y^{ij} measures how much more strongly channel i leads channel j than the other way around.

Independent sampling has no lag correlations, so $\Sigma_Y(\Pi) = C_0^Y$ and $\Gamma_Y(\Pi) = 0$.

Lemma 2.1 (Reversal parity)**.** *Any bounded $u, w : \mathsf{S} \to \mathbb{R}$ and $s \geq 0$ satisfy*

$$\mathbb{E}_{P^\dagger}[u(X_0)w(X_s)] = \mathbb{E}_P[u(X_s)w(X_0)].$$

In particular,

$$C_s^Y(P^\dagger) = C_s^Y(P)^\top, \qquad \Sigma_Y(P^\dagger) = \Sigma_Y(P), \qquad \Gamma_Y(P^\dagger) = -\Gamma_Y(P). \tag{2.6}$$

This is the path-reversal identity for stationary chains [24] — the relation $\pi(x)P^\dagger(x,y) = \pi(y)P(y,x)$ shows that reading a stationary path backwards exchanges the two source laws, and transposing the lag matrices gives (2.6).

So reversal keeps the symmetric part of each lag matrix and flips its antisymmetric part. The objective $L(\theta) = \sum_x \pi(x)\ell(\theta, x)$ and the autocovariance of any single channel are unchanged, which is why it takes at least two channels, such as a gradient and its square, to see the area. A reversible source has no area, but the converse fails for a fixed set of channels, since lags can cancel.

2.2 Losses and lag correlations

Let $c : \mathsf{S} \to \mathbb{R}$ have $\pi(c) = 0$ and $\sigma^2 = \pi(c^2) > 0$, and put $q = c^2 - \sigma^2$, so that $\pi(cq) = \pi(c^3)$. The sample and population losses are

$$\ell(\theta, x) = \frac{1}{2}(\theta + c(x))^2, \qquad L(\theta) = \frac{1}{2}\theta^2 + \frac{1}{2}\sigma^2, \tag{2.7}$$

so every kernel gives the same objective and the same minimizer, $\theta = 0$. The source statistics we need are the lag correlations between the gradient noise and its square,

$$C_s^{cq}(Q) = \mathbb{E}_Q[c(X_0)q(X_s)], \qquad C_s^{qc}(Q) = \mathbb{E}_Q[q(X_0)c(X_s)], \qquad s \geq 1, \tag{2.8}$$

where the gradient comes first in C_s^{cq} and the square comes first in C_s^{qc}. They are the off-diagonal entries of $C_s^Y(Q)$ for the channels $Y = (c, q)$. Independent sampling makes both lag correlations vanish. Under the source P, their antisymmetric and symmetric parts are

$$\gamma(s) = \frac{1}{2}\big(C_s^{cq}(P) - C_s^{qc}(P)\big), \qquad \chi(s) = \frac{1}{2}\big(C_s^{cq}(P) + C_s^{qc}(P)\big), \tag{2.9}$$

and the signed area between c and q is $\Gamma = \sum_{s\geq 1} \gamma(s)$. Reversal flips γ and keeps χ.

Fix $\delta > 0$ and set $d(v) = (v + \delta)^{-1/2}$. The smoothed denominator $d(v)$ replaces the usual $(\sqrt{v} + \epsilon)^{-1}$ so that the update is smooth at $v = 0$. We write $d_* = d(\sigma^2)$ and $d'_* = d'(\sigma^2) < 0$, so that $d'_*/d_* = -1/\{2(\sigma^2 + \delta)\}$.

Our results turn on the momentum-weighted lag sum of the introduction. For a momentum $\beta_1 \in [0, 1)$ it is

$$\mathcal{T}_Q = \pi(cq) + \sum_{s\geq 1}\big(C_s^{qc}(Q) + \beta_1^s C_s^{cq}(Q)\big). \tag{2.10}$$

The square-first correlations count fully, and the gradient-first ones carry the weight β_1^s that

momentum still gives the gradient. Independent sampling has $\mathcal{T}_\Pi = \pi(c^3)$.

Plain SGD on the same losses, $\theta_k = (1-h)\theta_{k-1} - hc(X_k)$, has stationary mean zero under every centered stationary source, although its stationary variance can change. So any displacement below comes from the adaptive denominator.

3 Adam with Fixed Memories

In practice β_1 and β_2 stay fixed while the learning rate is tuned. The memories then move fast compared with the parameter, so the parameter follows a drift averaged over their stationary law at each fixed parameter value. Temporal dependence changes that law, so it can move the equilibrium.

3.1 Frozen filters and averaging

Let the source have kernel $Q \in \{P, \Pi, P^\dagger\}$ and stationary law π, and let $c_k = c(X_k)$ with $\pi(c) = 0$ and $\sigma^2 = \pi(c^2) > 0$. Fix the memories $\beta_1, \beta_2 \in [0, 1)$, along with $\delta > 0$ and a weight decay $\lambda \geq 0$. The recursion is

$$\begin{aligned} g_k &= \theta_{k-1} + c_k, & m_k &= \beta_1 m_{k-1} + (1-\beta_1) g_k, \\ v_k &= \beta_2 v_{k-1} + (1-\beta_2) g_k^2, & \theta_k &= \theta_{k-1} - h\{d(v_k) m_k + \lambda \theta_{k-1}\}. \end{aligned} \tag{3.1}$$

This is Adam without bias correction, and $\beta_1 = 0$ gives RMSprop. To describe the memories at a fixed parameter value, we run three filters on the source alone,

$$\eta_k = \beta_1 \eta_{k-1} + (1-\beta_1) c_k, \qquad \xi_k = \beta_2 \xi_{k-1} + (1-\beta_2) c_k, \qquad \zeta_k = \beta_2 \zeta_{k-1} + (1-\beta_2) c_k^2, \tag{3.2}$$

and write $\Xi_k = (X_k, \eta_k, \xi_k, \zeta_k)$. We call (3.1) together with these filters the augmented recursion. With θ held fixed, the momentum is $\theta + \eta$ and the variance is $v_\theta = \theta^2 + 2\theta\xi + \zeta$, so the frozen update and its average are

$$\Phi(\theta, y) = (\theta + \eta)\, d(v_\theta(y)) + \lambda\theta, \qquad \bar{\Phi}_Q(\theta) = \int \Phi(\theta, y)\, \nu_Q(\mathrm{d}y), \tag{3.3}$$

where ν_Q is the unique stationary law of Ξ (Lemma A.1). It does not depend on θ. The parameter follows $\dot{\theta} = -\bar{\Phi}_Q(\theta)$ to leading order.

Assumption 3.1 (Stable averaged root)**.** The drift $\bar{\Phi}_Q$ has a zero θ_Q with $|\theta_Q| < 2\|c\|_\infty$, and for some $\omega_Q > 0$,

$$(\theta - \theta_Q)\bar{\Phi}_Q(\theta) \geq \omega_Q(\theta - \theta_Q)^2, \qquad |\theta| \leq 2\|c\|_\infty. \tag{3.4}$$

A zero always exists, because $\bar{\Phi}_Q(\pm 2\|c\|_\infty)$ have opposite signs, so the real content

is the stability inequality. It holds automatically when $\beta_1 = \beta_2$, since then $\eta = \xi$ and $\partial_\theta \Phi = (\zeta - \xi^2 + \delta)(v_\theta + \delta)^{-3/2} + \lambda > 0$. We call a stable zero θ_Q the frozen root. It is the equilibrium the parameter settles around. With unequal memories the inequality has to be checked in general, but two cases need no check at all.

Proposition 3.2 (Stability without a threshold)**.** *Fix $\beta_1, \beta_2 \in [0, 1)$, $\delta > 0$, and $\lambda \geq 0$. Suppose that c takes only two values, or that $\delta > \frac{1}{4}(\max c - \min c)^2$. Then every kernel $Q \in \{P, \Pi, P^\dagger\}$ satisfies Assumption 3.1, with a constant ω_Q that only depends on c, δ, and λ. The frozen root lies strictly between $-\max c$ and $-\min c$, and on that interval $\Phi(\cdot, y)$ is increasing for ν_Q-almost every filter state y.*

The interval is the range of the per-sample minimizers $-c(x)$. Inside it, the variance held in memory and the smoothing term together control the mismatch between the two memories, and when c takes two values the variance alone cancels the worst possible mismatch. Outside the interval every sample pushes the parameter back. So the theorem below covers every two-state source at any β_1 and β_2, including the family of Section 4.1. The proof is in Appendix B.4.

Theorem 3.3 (Fixed-memory averaging)**.** *Fix $Q, \beta_1, \beta_2, \delta, \lambda$, assume Assumption 3.1, and let h be small enough. The augmented recursion has a compact forward-invariant class $\mathcal{A}_h$, which contains every start with $m_0 = v_0 = 0$ and $|\theta_0| \leq \|c\|_\infty$, and an invariant law on it. Every invariant law on $\mathcal{A}_h$ satisfies*

$$\mathbb{E}[|\theta - \theta_Q|^2] \leq C_Q h, \qquad \mathbb{E}[|\theta - \theta_Q|^4] \leq C_Q h^2, \qquad |\mathbb{E}[\theta] - \theta_Q| \leq C_Q h, \tag{3.5}$$

and the same bounds hold in limsup, in expectation and almost surely, for long-run averages from any start in $\mathcal{A}_h$. The constants are uniform over $\beta_1, \beta_2 \leq \bar{\beta} < 1$ when ω_Q is bounded below.

Averaging with fast Markov memories is classical [2, 22]. What we need is a stationary version with moment bounds that hold for every invariant law. The proof in Appendix B reduces the recursion to θ alone with an $O(h^2)$ tracking error and then runs a moment argument on the reduced recursion. The long-run statement carries over to the stationary estimates in Theorem 3.4 and Corollary 3.5, with the same proof. If the source and independent sampling both have stable roots, the dependence shift is therefore $\theta_P - \theta_\Pi + O(h)$. It has no reason to vanish as $h \to 0$.

3.2 Expansion of the frozen root

The roots are hard to compute in general, but they have an explicit expansion to first order in $1 - \beta_2$. Fix $\delta > 0$, $\lambda \geq 0$, and $\beta_1 \in [0, \bar{\beta}]$ with $\bar{\beta} < 1$, write $\beta_2 = 1 - \alpha$, and write $\theta_{Q,\alpha}$ for

the frozen root at this α. Its leading term is set by the momentum-weighted lag sum $\mathcal{T}_Q$ of (2.10).

Theorem 3.4 (Frozen roots near $\beta_2 = 1$)**.** *There is $\alpha_0 > 0$ such that, for $0 < \alpha \le \alpha_0$, the kernels P, Π, and $P^\dagger$ all satisfy Assumption 3.1 with a stability constant independent of α, β_1, and the kernel, and*

$$\boxed{\theta_{Q,\alpha} = -\frac{\alpha d'_*}{d_* + \lambda}\,\mathcal{T}_Q + O(\alpha^2),} \tag{3.6}$$

uniformly in β_1. At each such α and all small h, any invariant law μ_h^Q on the class of Theorem 3.3 satisfies

$$\mathbb{E}_{\mu_h^Q}[\theta] = \theta_{Q,\alpha} + O_\alpha(h), \tag{3.7}$$

where $O_\alpha(h)$ is bounded by $C_\alpha h$ for $h \le h_0(\alpha)$.

The proof is in Appendix B.3. The lag weights of (2.10) set the leading root directly. The parameter barely moves while the memories turn over, so the root only depends on the source and the two memories, and not on the learning rate.

Independent sampling has $\mathcal{T}_\Pi = \pi(c^3)$, so its root is displaced whenever the gradient noise is skewed, by the same amount for every β_1 at this order. Subtracting the two roots gives

$$\theta_{P,\alpha} - \theta_{\Pi,\alpha} = -\frac{\alpha d'_*}{d_* + \lambda} \sum_{s \ge 1} \big(C_s^{qc}(P) + \beta_1^s C_s^{cq}(P) \big) + O(\alpha^2), \tag{3.8}$$

and by (3.7) this is the dependence shift of the stationary means up to $O_\alpha(h)$. Reversal swaps the two orders, so $\mathcal{T}_P = \mathcal{E}_{\beta_1} - \Gamma_{\beta_1}$ and $\mathcal{T}_{P^\dagger} = \mathcal{E}_{\beta_1} + \Gamma_{\beta_1}$, with the even coefficient and the momentum-weighted area

$$\mathcal{E}_{\beta_1} = \pi(cq) + \sum_{s \ge 1} (1 + \beta_1^s)\chi(s), \qquad \Gamma_{\beta_1} = \sum_{s \ge 1} (1 - \beta_1^s)\gamma(s). \tag{3.9}$$

The shared part of the two roots is $-\alpha d'_* \mathcal{E}_{\beta_1}/(d_* + \lambda) + O(\alpha^2)$, and the orientation component is

$$\frac{\theta_{P,\alpha} - \theta_{P^\dagger,\alpha}}{2} = \frac{\alpha d'_*}{d_* + \lambda}\,\Gamma_{\beta_1} + O(\alpha^2). \tag{3.10}$$

Reversal swaps the weights 1 and β_1^s, which leaves $1 - \beta_1^s$. Intuitively, momentum still carries the gradient from s steps ago with weight β_1^s, so a gradient that came before a burst of large squared gradients is still partly in the update after the burst, where it counts with the opposite orientation. At $\beta_1 = 0$ the weight is one and $\Gamma_0 = \Gamma$. Signed lags can cancel, so momentum does not shrink every area.

The two estimates of Theorem 3.4 are an iterated limit — first $h \to 0$ at fixed α, then $\alpha \to 0$. We do not claim a bound that is uniform in both, since C_α and $h_0(\alpha)$ may deteriorate as $\alpha \to 0$. Learning-rate schedules within a run are a different scaling, which we do not

study.

3.3 Dependence can lower or raise the loss

With fixed memories the stationary loss tends to the loss at the frozen root as $h \to 0$, so the root expansion decides whether dependence lowers or raises it.

Corollary 3.5 (Loss under changes of source)**.** *Under Theorem 3.4, any two kernels* $Q, Q' \in \{P, \Pi, P^\dagger\}$ *satisfy*

$$L(\theta_{Q,\alpha}) - L(\theta_{Q',\alpha}) = \frac{\alpha^2}{2}\Big(\frac{d'_*}{d_* + \lambda}\Big)^2(\mathcal{T}_Q^2 - \mathcal{T}_{Q'}^2) + O(\alpha^3). \tag{3.11}$$

In particular, since $\mathcal{T}_\Pi = \pi(c^3)$,

$$L(\theta_{P,\alpha}) - L(\theta_{\Pi,\alpha}) = \frac{\alpha^2}{2}\Big(\frac{d'_*}{d_* + \lambda}\Big)^2(\mathcal{T}_P^2 - \pi(c^3)^2) + O(\alpha^3), \tag{3.12}$$

whereas reversal gives

$$L(\theta_{P,\alpha}) - L(\theta_{P^\dagger,\alpha}) = -2\alpha^2\Big(\frac{d'_*}{d_* + \lambda}\Big)^2\mathcal{E}_{\beta_1}\Gamma_{\beta_1} + O(\alpha^3). \tag{3.13}$$

At each such α *and all small* h*, the stationary expected loss differences equal the corresponding root-loss differences up to* $O_\alpha(h)$*. If* $\mathcal{T}_P^2 \neq \pi(c^3)^2$*, dependence gives a lower loss than independent sampling exactly when* $|\mathcal{T}_P| < |\pi(c^3)|$*, once* α *and then* h *are small enough. If* $\mathcal{E}_{\beta_1}\Gamma_{\beta_1} \neq 0$*, the forward source gives a lower loss than its reversal exactly when this product is positive, with the same order of limits.*

The proof is in Appendix B.3. The sign depends on the source. In the two-state family of Section 4.1, positive dependence raises the loss and negative dependence lowers it. The asymmetric cycle of Appendix C.3 shows both signs at once, since its forward source gives a lower loss than independent sampling and its reversal a higher one. Under reversal, the root losses differ by twice the product of the average root and half the root difference, and either factor can vanish, as in the symmetric cycle of Section 4.2. Two states are enough for dependence, but orientation needs at least three, because every stationary two-state chain is reversible.

3.4 Additive noise in any dimension

Adam's memories work coordinate by coordinate, and all they see is the gradient. Freeze the parameter, and the gradient in each coordinate is a constant plus noise, exactly as in the scalar model. So the frozen roots above already describe the equilibrium of a much larger class of problems, as long as the gradient noise does not depend on the parameter.

Let $\theta \in \mathbb{R}^n$, and let the sample losses have gradients

$$\nabla_\theta \ell(\theta, x) = \nabla L(\theta) + c(x), \qquad c : \mathsf{S} \to \mathbb{R}^n, \qquad \pi(c) = 0, \tag{3.14}$$

where L is twice continuously differentiable and strongly convex, and no coordinate c_i is constant. Run (3.1) in each coordinate with $\lambda = 0$, so that coordinate i has $g_k^i = \partial_i L(\theta_{k-1}) + c_i(X_k)$ and its own memories m_k^i and v_k^i. Let $\bar{\Phi}_Q^{(i)}$ be the scalar drift (3.3) built from the channel c_i, with $\lambda = 0$. If θ is held fixed, coordinate i is the scalar frozen system at the parameter value $\partial_i L(\theta)$, so the averaged drift of the n-dimensional recursion is

$$\bar{\Phi}_Q(\theta)_i = \bar{\Phi}_Q^{(i)}\big(\partial_i L(\theta)\big), \qquad i = 1, \dots, n. \tag{3.15}$$

Proposition 3.6 (Additive noise)**.** *Suppose that every channel c_i satisfies Assumption 3.1, with frozen root $\theta_Q^{(i)}$ and constant $\omega_Q^{(i)}$, and put $r_Q = (\theta_Q^{(1)}, \dots, \theta_Q^{(n)})$ and $\omega = \min_i \omega_Q^{(i)}$. Then the averaged drift (3.15) has exactly one zero θ_Q, the solution of*

$$\nabla L(\theta_Q) = r_Q. \tag{3.16}$$

The function $D(\theta) = L(\theta) - L(\theta_Q) - \langle r_Q, \theta - \theta_Q \rangle$ is positive away from θ_Q and decreases strictly along the averaged flow $\dot{\theta} = -\bar{\Phi}_Q(\theta)$, with

$$\frac{\mathrm{d}}{\mathrm{d}t} D(\theta) \le -\omega |\nabla L(\theta) - r_Q|^2 \qquad \text{wherever } |\partial_i L(\theta)| \le 2\|c_i\|_\infty \text{ for every } i. \tag{3.17}$$

So θ_Q attracts every solution of the averaged flow. For the quadratic $L(\theta) = \frac{1}{2}(\theta - \theta_*)^\top A(\theta - \theta_*)$ with A symmetric and positive definite,

$$\theta_Q = \theta_* + A^{-1} r_Q, \qquad L(\theta_Q) - L(\theta_*) = \tfrac{1}{2}\, r_Q^\top A^{-1} r_Q. \tag{3.18}$$

The equilibrium is the point where the population gradient equals r_Q, and it is not the minimizer unless $r_Q = 0$. The loss only enters through this one equation, and the noise enters coordinate by coordinate. Theorem 3.4 applies to each channel, so for small α

$$(r_Q)_i = \frac{\alpha}{2(\sigma_i^2 + \delta)}\, \mathcal{T}_Q^{(i)} + O(\alpha^2), \tag{3.19}$$

where $\sigma_i^2 = \pi(c_i^2)$ and $\mathcal{T}_Q^{(i)}$ is the lag sum (2.10) of the channel c_i. In the quadratic case the displacement is $A^{-1} r_Q$, so directions with small curvature are displaced the most. Proposition 3.2 supplies the assumption when each c_i takes two values or δ is large enough. The proof is in Appendix B.5.

The proposition is about the averaged drift and its flow. Carrying Theorem 3.3 over to the coupled recursion needs a compact forward-invariant class, which the scalar proof gets

from the linearity of the gradient, and we do not pursue it here.

4 Examples and Numerical Evidence

We work out two examples. A reversible two-state source shows that dependence alone can lower or raise the loss, and a symmetric three-state cycle isolates orientation at matched covariance. Appendix C has the computations behind them, and an asymmetric version of the cycle whose two orientations end at different losses.

4.1 A reversible two-state source

Take two states with

$$c = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \qquad \pi = \Big(\frac{2}{3}, \frac{1}{3}\Big), \qquad P_\rho = \rho I + (1-\rho)\Pi, \qquad -\tfrac{1}{2} < \rho < 1, \tag{4.1}$$

so $\sigma^2 = 2$ and $\pi(c^3) = -2$. When $\rho > 0$ the source is sticky, repeating its last state with probability ρ and otherwise drawing a fresh one from π. A negative ρ still gives a positive stochastic matrix, and then the source switches states more often than independent sampling. Every member of the family is reversible, so every area vanishes. Centered functions satisfy $P_\rho^s f = \rho^s f$, so both lag correlations equal $\rho^s \pi(c^3)$, and

$$\mathcal{T}_{P_\rho} = \pi(c^3)\Big(\frac{1}{1-\rho} + \frac{\beta_1\rho}{1-\beta_1\rho}\Big). \tag{4.2}$$

Theorem 3.4 then gives

$$\frac{\theta_{P_\rho,\alpha}}{\theta_{\Pi,\alpha}} = \frac{1}{1-\rho} + \frac{\beta_1\rho}{1-\beta_1\rho} + O(\alpha). \tag{4.3}$$

At $\beta_1 = 0$, which is RMSprop, the multiplier is $1/(1-\rho)$. The corresponding ratio of excess losses is

$$\frac{L(\theta_{P_\rho,\alpha}) - L(0)}{L(\theta_{\Pi,\alpha}) - L(0)} = \Big(\frac{1}{1-\rho} + \frac{\beta_1\rho}{1-\beta_1\rho}\Big)^2 + O(\alpha). \tag{4.4}$$

The multiplier in (4.3) is positive, strictly increasing in ρ, and equal to one at independence. So negative dependence lowers the limiting loss and positive dependence raises it. At $\beta_1 = 1/2$ the limiting excess-loss ratios are $961/2025 \approx 0.475$ for $\rho = -1/4$ and $49/9 \approx 5.44$ for $\rho = 1/2$. Both effects come from the same marginal law and involve no orientation at all. By Corollary 3.5 they carry over to the stationary expected losses for each fixed nonzero ρ, once α and then h are small enough. Proposition 3.2 gives the stable root for every member of the family at any β_1 and β_2, so Theorem 3.3 needs no smallness of α here. The expansion (4.3) is a different matter. It holds for a fixed ρ as $\alpha \to 0$ and needs a run to be short compared with the variance memory, $\alpha \ll 1-\rho$. It cannot hold as $\rho \uparrow 1$, where the

multiplier grows without bound while the root stays in $[-1, 2]$.

The simulations use the sticky source $\rho = 1/2$ with $\delta = 1$ and $\lambda = 0$ and check the mean shifts (Figure 1), and Appendix C gives the predicted coefficients.

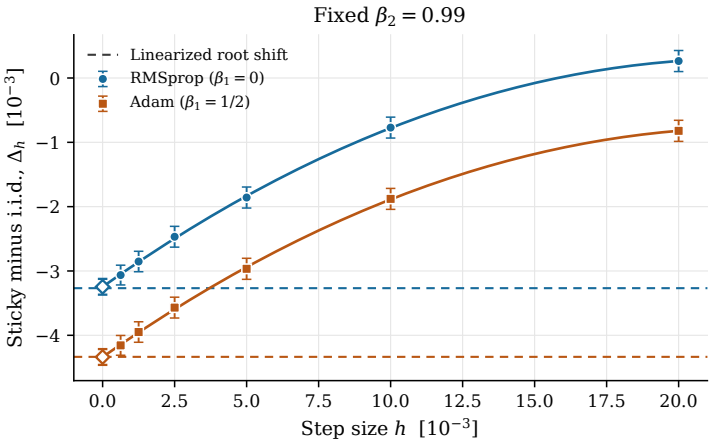


Figure 1: The sticky source ($\rho = 1/2$) against independent sampling at $\beta_2 = 0.99$, with Δ_h the difference of their mean parameters (Section 4.3). Points are simulated differences with quadratic fits, and dashed lines are the linearized root differences of Table 1. Open diamonds are the fitted intercepts at $h = 0$, and bars show two Monte Carlo standard errors.

4.2 A symmetric cycle

The second example is a lazy three-state cycle with $\delta = 2/3$. Take

$$P = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \pi = \frac{1}{3}(1, 1, 1), \qquad c = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \qquad q = \frac{1}{3}\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}. \tag{4.5}$$

Here $P^\dagger = P^\top$ and $\sigma^2 = 2/3$. The forward chain climbs through the values $-1, 0, 1$ and then drops back to -1, and at each step it either stays put or moves on, with probability $1/2$ each. Each visit to -1 starts right after the large gradient at 1, so RMSprop damps the steps there, which move the parameter up. Each visit to 1 starts after the calm middle state, where $c = 0$, so the steps there, which move the parameter down, are damped less. The downward steps win, and RMSprop settles below the minimizer. The reversed chain runs the other way and settles above it.

All three kernels P, $P^\dagger$, and Π have the same long-run covariance for every vector of centered observables of the source (Appendix C). Yet their areas are $\Gamma = 2/9$, $-2/9$, and 0. The symmetric parts vanish too ($\pi(c^3) = 0$ and $\chi(s) = 0$ by Lemma C.1), so independent sampling sits at the minimizer to first order and the whole dependence shift is the orientation component. Swapping the states where $c = -1$ and $c = 1$ turns P into $P^\dagger$ and c into $-c$, so

the two orientations are mirror images and $\theta_{P^\dagger,\alpha} = -\theta_{P,\alpha}$ (Lemma C.1). Then $\mathcal{E}_{\beta_1} = 0$, the two root losses are equal, and with $d'_*/d_* = -3/8$ and $\lambda = 0$ Theorem 3.4 gives

$$\theta_{P,\alpha} = -\frac{\alpha}{12}\,\frac{(1-\beta_1)(4-\beta_1)}{\beta_1^2 - 2\beta_1 + 4} + O(\alpha^2), \tag{4.6}$$

since $\Gamma_{\beta_1} = \frac{2}{9}(1-\beta_1)(4-\beta_1)/(\beta_1^2 - 2\beta_1 + 4)$. The coefficient is $-1/12$ for RMSprop, $-7/156$ at $\beta_1 = 1/2$, and $-31/3612$ at $\beta_1 = 9/10$, and weight decay multiplies it by $d_*/(d_* + \lambda)$. The negative sign matches the story above. Momentum only shrinks the effect here, and the coefficient tends to zero as $\beta_1 \uparrow 1$.

4.3 Simulations

The simulations check the predicted mean shifts with two paired comparisons. The sticky and i.i.d. streams share their fresh draws from π. The i.i.d. stream uses every draw, while the sticky stream keeps its last state with probability ρ and otherwise takes the fresh draw. On the symmetric cycle, a stationary P-path read backwards is a stationary $P^\dagger$-path (Lemma 2.1), so the two optimizers read the central segment of one path in opposite directions, after separate burn-ins. These pairings reduce sampling noise. We write Δ_h for the difference of the window means, averaged over replicates. It is the full sticky-minus-i.i.d. difference in the first comparison and half the forward-minus-reversed difference in the second, and Table 1 reports its weighted least-squares extrapolation to $h = 0$. The sticky runs use $\rho = 1/2$ and $\delta = 1$, the cycle runs use $\delta = 2/3$, and all runs use $\beta_2 = 0.99$.

Quantity	Setting	Extrapolated	Predicted
Sticky minus i.i.d.: full difference			
Δ_h ($\times 10^3$), RMSprop	$\beta_1 = 0$	-3.25 ± 0.06	-3.27
Δ_h ($\times 10^3$), Adam	$\beta_1 = 0.5$	-4.34 ± 0.06	-4.34
Symmetric cycle: half the forward-minus-reversed difference			
Δ_h ($\times 10^5$), Adam	$\beta_1 = 0.9$	-6.98 ± 0.17	-7.11
Δ_h ($\times 10^5$), Adam	$\beta_1 = 0.5$	-43.62 ± 0.17	-43.60

Table 1: Extrapolated mean shifts at $\beta_2 = 0.99$ against predictions. The predictions are the full or half differences of the linearized roots $-d'_*\mathbb{E}_{\nu_Q}[\eta(\zeta - \sigma^2)]/(d_* + \lambda)$, evaluated with the exact filtered covariance of Lemma B.3. Each $\pm$ is one reported Monte Carlo standard error, excluding extrapolation and root-linearization error.

Both comparisons keep nonzero fitted intercepts consistent with the linearized root differences, as expected when the shift survives as $h \to 0$ (Figures 1 and 2). The linearized roots differ from the leading small-α terms by 2–3% in most rows and by about 17% on the symmetric cycle at $\beta_1 = 0.9$ (Figure 3). Proposition 3.2 covers the sticky runs, so Theorem 3.3 applies to them at $\beta_2 = 0.99$. For the cycle the stability threshold of Theorem 3.4 is not computed, so the theorem does not certify $\beta_2 = 0.99$ there. The sticky curves bend at

larger h, because for $h \gtrsim 0.01$ the parameter relaxes about as fast as the variance memory. Their intercepts therefore use quadratic fits over six step sizes down to $h = 0.000625$, and Appendix D gives the fit sensitivity and the i.i.d. baselines.

We also repeated the cycle runs with the usual denominator $d(v) = (\sqrt{v}+\epsilon)^{-1}$ at $\epsilon = \sqrt{2/3}$ (Figure 2). Our proofs do not cover this denominator. At $\beta_2 = 0.99$, the extrapolated half-differences are $(-43.52 \pm 0.20) \times 10^{-5}$ for $\beta_1 = 1/2$ and $(-6.87 \pm 0.21) \times 10^{-5}$ for $\beta_1 = 9/10$, with one Monte Carlo standard error. The usual denominator has the same ratio $d'_*/d_* = -1/\{2\sigma(\sigma+\epsilon)\} = -3/8$ as the smoothed one, so the linearized values in Table 1, -43.60×10^{-5} and -7.11×10^{-5}, apply to both. Its fits lie within 1.2 reported standard errors of these values. Their magnitudes are about 3% and 20% below the leading small-α predictions, -44.87×10^{-5} and -8.58×10^{-5}, much as in the smoothed runs. The Monte Carlo errors exclude extrapolation and approximation errors. The loss effects of Corollary 3.5 and the asymmetric cycle are analytic results here, and we have not simulated them.

5 Conclusion

Adaptive optimizers read the order of their data. Temporal dependence enters RMSprop and Adam through a momentum-weighted lag sum, even when the objective and the distribution of each sample are held fixed. The sum counts a squared gradient fully when it comes before a gradient and only through momentum when it comes after. With fixed memories it sets the displaced equilibrium to first order in $1 - \beta_2$, and the displacement does not vanish with the learning rate.

The loss follows the frozen root, so dependence alone can lower or raise it, even for a reversible source. Reversal keeps the long-run covariance, so it isolates the extra effect of orientation, and the symmetric cycle shows that matching independent sampling in the marginal law and the long-run covariance does not pin down where Adam settles.

Our stationary results are for scalar quadratic losses with a smoothed denominator, and the root expansion comes with a threshold for β_2 that is not explicit. Two-state sources need no threshold for the averaging step, and the averaged equilibrium extends to strongly convex losses with additive noise in any dimension, but for that coupled recursion the averaging step is not proved here. Natural next steps are that step, the usual denominator, noise that depends on the parameter, and sources whose runs are as long as the variance memory, where the lag sum is no longer a small correction. Within our model, the marginal law and the long-run covariance of the data leave out information that can move where an adaptive optimizer settles.

Code, data, and disclosure

The accompanying archive, `focused_reproducibility.zip`, contains the simulation code, stored run summaries, deterministic root calculations, and scripts for the fits and figures. Its README records the protocols and their qualifications. The sticky-source runs only kept aggregate means and reported standard errors, so the archive reproduces those fits but cannot rebuild the standard errors from the replicates.

The author used OpenAI Codex, ChatGPT, and Anthropic Claude for editing, proof checks, numerical code and figures, and references. The framing around an i.i.d. baseline, the sticky-source example, and the asymmetric loss-gap example were suggested by ChatGPT. The closed-form family in β_1, the stability proposition (Proposition 3.2), and the additive-noise proposition (Proposition 3.6) were developed with Claude. Codex also helped formulate the source-to-source loss comparisons, check the stability and additive-noise arguments, revise the dependence-based exposition, and regenerate the fixed-memory figures. The author is responsible for the manuscript and for independently verifying its arguments, computations, and citations.

A Two tools

The proofs use two standard facts: a mixing bound for filters driven by the source, and the invariance of limiting empirical measures. Throughout the appendices $\bar{\beta} < 1$ bounds the memories, and generic constants may depend on the fixed model.

Lemma A.1 (Mixing of causal filters)**.** *Let $\Xi_k = (X_k, y_k)$, where X is the source with kernel $Q \in \{P, \Pi, P^\dagger\}$ and $y_k = \Lambda y_{k-1} + \phi(X_k)$ for a diagonal Λ with entries in $[0, \bar{\beta}]$ and a vector-valued function ϕ. Equip the compact filter range with the metric $\mathbf{1}_{x \neq x'} + |y - y'|$. Then Ξ, with transition operator $\mathcal{K}_Q$, has a unique invariant law ν_Q, and there is $C < \infty$, uniform over the three kernels and over Λ, such that every bounded Lipschitz f with $\nu_Q(f) = 0$ satisfies*

$$\|\mathcal{K}_Q^n f\|_\infty \leq C e^{-n/C} \|f\|_{\mathrm{BL}}.$$

Centered smooth families $f(z, \cdot)$ with bounded Lipschitz derivatives up to order three have Poisson correctors $\sum_{n \geq 0} \mathcal{K}_Q^n f$ with the same derivative bounds.

Proof. This is the contraction argument for iterated random functions [11]. Couple two copies of the source so that their meeting time has an exponential tail. After they meet, the two filter states contract by a factor of at most $\bar{\beta}$ per step. Splitting on whether they have met by time $\lfloor n/2 \rfloor$ gives the bound, and integrating one start against an invariant law gives uniqueness. Because ν_Q does not depend on z, differentiating a centered family keeps it centered, and the geometric bound lets us differentiate the Poisson series term by term. □

Since Ξ does not depend on the parameter, its marginal under every invariant law of the augmented recursion is ν_Q. The filtered kernels for P and $P^\dagger$ are not time reversals of each other: reversing a causal filter would make it read the future, while (3.2) stays causal under every kernel. So the frozen roots of the two orientations are computed separately and compared at the end.

The long-run statements in the paper rest on one standard fact about compact Feller chains.

Lemma A.2 (Pathwise averages)**.** *Let (Ω_k) be a Markov chain on a compact metric space $\mathcal{X}$ whose kernel $\mathcal{K}$ maps continuous functions to continuous functions. Almost surely, every weak limit point of the empirical measures $n^{-1} \sum_{k<n} \delta_{\Omega_k}$ is invariant for $\mathcal{K}$.*

Proof. For continuous f, the sum $\sum_{k<n} \{f(\Omega_{k+1}) - \mathcal{K}f(\Omega_k)\}$ is a martingale with increments bounded by $2\|f\|_\infty$, so it is $o(n)$ almost surely by Azuma's inequality and the Borel–Cantelli lemma. Since $C(\mathcal{X})$ is separable, this holds for a countable dense set of f at once. Shifting the first sum by one step changes it by at most $2\|f\|_\infty$, so every weak limit point ν of the empirical measures satisfies $\nu(f) = \nu(\mathcal{K}f)$ for these f, and density gives $\nu\mathcal{K} = \nu$. □

B Fixed-memory Adam

B.1 Reduction to the parameter

Define the tracking errors and the step

$$p_k = m_k - \theta_k - \eta_k, \qquad \tau_k = v_k - v_{\theta_k}(\Xi_k), \qquad \Psi_k = d(v_k)m_k + \lambda\theta_{k-1}. \tag{B.1}$$

Lemma B.1 (Exact tracking identities)**.** *Along every source path,*

$$\begin{aligned} &m_k = \theta_{k-1} + \eta_k + \beta_1 p_{k-1}, \quad p_k = \beta_1 p_{k-1} + h\Psi_k, && \theta_k = \theta_{k-1} - h\Psi_k, \\ &v_k = v_{\theta_{k-1}}(\Xi_k) + \beta_2\tau_{k-1}, \quad \tau_k = \beta_2\tau_{k-1} + h\Psi_k(\theta_{k-1} + \theta_k + 2\xi_k). \end{aligned} \tag{B.2}$$

Proof. Substitute $m_{k-1} = \theta_{k-1} + \eta_{k-1} + p_{k-1}$ into the momentum update, and use $v_\theta(\Xi_k) = \beta_2 v_\theta(\Xi_{k-1}) + (1-\beta_2)(\theta + c_k)^2$ together with $v_{\theta_{k-1}}(\Xi_k) - v_{\theta_k}(\Xi_k) = h\Psi_k(\theta_{k-1} + \theta_k + 2\xi_k)$. □

The filters stay in $\mathcal{F} = \{|\eta| \le \|c\|_\infty,\ |\xi| \le \|c\|_\infty,\ \xi^2 \le \zeta \le \|c\|_\infty^2\}$, so $v_\theta \ge 0$. Put $\bar\theta = 2\|c\|_\infty$, $\bar v = (\bar\theta + \|c\|_\infty)^2$, $\bar d = \delta^{-1/2}$, $\bar\Psi = \bar d(\bar\theta + 2\|c\|_\infty) + \lambda\bar\theta$, $\bar p = 2\bar\Psi/(1-\beta_1)$, and $\bar\tau = 2\bar\Psi(2\bar\theta + 2\|c\|_\infty)/(1-\beta_2)$, and take h with $h(\bar d + \lambda) \le 1$ and $\beta_1\bar p h \le \bar\theta - \|c\|_\infty$. The class is

$$\mathcal{A}_h = \{|\theta| \le \bar\theta,\ 0 \le v \le \bar v,\ \Xi \in \mathcal{F},\ |p| \le \bar p h,\ |\tau| \le \bar\tau h\}. \tag{B.3}$$

Lemma B.2 (Confinement and reduction)**.** *The class $\mathcal{A}_h$ is compact and forward invariant along every source path, it carries an invariant law, and on it*

$$\theta_k = \theta_{k-1} - h\Phi(\theta_{k-1}, \Xi_k) + h^2\mathcal{R}_{h,k}, \qquad |\mathcal{R}_{h,k}| \le C\Big(\frac{1}{1-\beta_1} + \frac{1}{1-\beta_2}\Big). \tag{B.4}$$

A start with $m_0 = v_0 = 0$ and $|\theta_0| \le \|c\|_\infty$, augmented by $\eta_0 = \xi_0 = -\theta_0$ and $\zeta_0 = \theta_0^2$, lies in $\mathcal{A}_h$.

Proof. The incoming bound $|\theta_{k-1}| \le \bar\theta$ gives $0 \le v_k \le \bar v$ and $|m_k| \le \bar\theta + 2\|c\|_\infty$, so $|\Psi_k| \le \bar\Psi$ and $|p_k| \le \beta_1\bar p h + h\bar\Psi \le \bar p h$. Writing $\theta_k = \{1 - h[d(v_k) + \lambda]\}\theta_{k-1} - hd(v_k)(\eta_k + \beta_1 p_{k-1})$ with a nonnegative first coefficient and $|\eta_k + \beta_1 p_{k-1}| \le \bar\theta$ gives $|\theta_k| \le \bar\theta$. Only then do we use the last tracking identity, which gives $|\tau_k| \le \beta_2\bar\tau h + h\bar\Psi(2\bar\theta + 2\|c\|_\infty) \le \bar\tau h$, so the argument is not circular. The first two identities give $m_k = \theta_{k-1} + \eta_k + O(h/(1-\beta_1))$ and $v_k = v_{\theta_{k-1}}(\Xi_k) + O(h/(1-\beta_2))$, which gives (B.4). The augmented start has $p_0 = \tau_0 = 0$. The sample maps are continuous, so the joint chain is Feller on a compact space, and Cesàro averages of its laws have invariant limit points. □

B.2 Proof of Theorem 3.3

Lemma A.1, applied to (η, ξ, ζ) with $\Lambda = \mathrm{diag}(\beta_1, \beta_2, \beta_2)$, gives the unique law ν_Q and Poisson correctors $\widehat N_Q$ and $\psi_Q = \mathcal{K}_Q\widehat N_Q$ for the centered field $N_Q(\theta, y) = \bar\Phi_Q(\theta) - \Phi(\theta, y)$, with three bounded θ-derivatives. By (B.4),

$$\theta_k - \theta_{k-1} = h\{-\bar\Phi_Q(\theta_{k-1}) + N_Q(\theta_{k-1}, \Xi_k)\} + h^2\mathcal{R}_{h,k}. \tag{B.5}$$

Moments. Write $\Delta_k = \theta_k - \theta_Q$ and let $\mathcal{G}_k$ be the history up to time k. The Poisson identity

$$N_Q(\theta_{k-1}, \Xi_k) = \mathcal{M}_k + \psi_Q(\theta_{k-1}, \Xi_{k-1}) - \psi_Q(\theta_{k-1}, \Xi_k),$$
$$\mathcal{M}_k = \widehat{N}_Q(\theta_{k-1}, \Xi_k) - \psi_Q(\theta_{k-1}, \Xi_{k-1}),$$

splits the noise into a martingale difference and a coboundary. We correct Δ^2 and Δ^4 by the coboundary,

$$\widetilde{V}_k = \Delta_k^2 + 2h\Delta_k\psi_Q(\theta_k, \Xi_k), \qquad \widetilde{V}_{2,k} = \Delta_k^4 + 4h\Delta_k^3\psi_Q(\theta_k, \Xi_k).$$

The source jump in the correction cancels the coboundary exactly, and only the change of state $h\{g(\theta_k, \Xi_k) - g(\theta_{k-1}, \Xi_k)\}$ remains, with $g = 2\Delta\psi_Q$ or $g = 4\Delta^3\psi_Q$. Taylor expansion, the bounded increments $|\theta_k - \theta_{k-1}| \le Ch$, and (3.4) then give

$$\mathbb{E}[\widetilde{V}_k - \widetilde{V}_{k-1} \mid \mathcal{G}_{k-1}] \le -2\omega_Q h\Delta_{k-1}^2 + Ch^2, \tag{B.6}$$
$$\mathbb{E}[\widetilde{V}_{2,k} - \widetilde{V}_{2,k-1} \mid \mathcal{G}_{k-1}] \le -4\omega_Q h\Delta_{k-1}^4 + Ch^2\Delta_{k-1}^2 + Ch^3|\Delta_{k-1}| + Ch^4. \tag{B.7}$$

For the second bound, $g = 4\Delta^3\psi_Q$ has θ-derivatives of order Δ^2 and $|\Delta|$ on the compact class, and the $h^2\mathcal{R}_{h,k}$ term obeys the same bounds. Under an invariant law both corrected increments have mean zero, so (B.6) gives $\mathbb{E}[\Delta^2] \le Ch$, and then (B.7) gives $\mathbb{E}[\Delta^4] \le Ch^2$, with constants that do not depend on the invariant law. The corrected functions need not be nonnegative, since boundedness is enough here.

The mean. Summing the Poisson identity, with $\psi_Q(\theta_k, \Xi_k)$ inserted, gives

$$\Big|\frac{1}{n}\sum_{k=1}^{n}\mathbb{E}[N_Q(\theta_{k-1}, \Xi_k)]\Big| \le Ch + \frac{C}{n} \tag{B.8}$$

from any start in $\mathcal{A}_h$. Under an invariant law, the pair (θ_{k-1}, Ξ_k) has a fixed distribution, so $|\mathbb{E}[N_Q]| \le Ch$, and the stationary mean of (B.5) gives $|\mathbb{E}[\bar{\Phi}_Q(\theta)]| \le Ch$. Differentiating under ν_Q,

$$\bar{\Phi}_Q'(\theta) = \mathbb{E}_{\nu_Q}[d(v_\theta) + 2(\theta+\eta)(\theta+\xi)d'(v_\theta)] + \lambda, \tag{B.9}$$
$$\bar{\Phi}_Q''(\theta) = \mathbb{E}_{\nu_Q}[(6\theta + 4\xi + 2\eta)d'(v_\theta) + 4(\theta+\eta)(\theta+\xi)^2 d''(v_\theta)], \tag{B.10}$$

and both are bounded on $[-\bar{\theta}, \bar{\theta}]$. So $\bar{\Phi}_Q(\theta) = \bar{\Phi}_Q'(\theta_Q)(\theta - \theta_Q) + O((\theta - \theta_Q)^2)$ with $\bar{\Phi}_Q'(\theta_Q) \ge \omega_Q$, and the second moment gives $|\mathbb{E}[\theta] - \theta_Q| \le Ch$. Long-run averages follow because every limit point of the averaged laws is invariant on the compact Feller class, and almost surely so is every limit point of the empirical measures along a path (Lemma A.2). □

B.3 Expansion of the frozen root

Throughout, $\beta_2 = 1 - \alpha$, $u_\alpha = \zeta - \sigma^2$, and the bounds are uniform over $Q \in \{P, \Pi, P^\dagger\}$ and $\beta_1 \le \bar{\beta}$, with constants that may depend on the fixed source.

Lemma B.3 (Exact filtered covariance). *For $0 < \alpha < 1$ and each kernel Q,*

$$\mathbb{E}_{\nu_Q}[\eta u_\alpha] = \frac{(1-\beta_1)\alpha}{1-\beta_1\beta_2}\Big[\pi(cq) + \sum_{s\geq 1}\beta_1^s C_s^{cq}(Q) + \sum_{s\geq 1}\beta_2^s C_s^{qc}(Q)\Big], \tag{B.11}$$

and as $\alpha \downarrow 0$,

$$\mathbb{E}_{\nu_Q}[\eta u_\alpha] = \alpha\mathcal{T}_Q + O(\alpha^2). \tag{B.12}$$

Proof. On a stationary two-sided path, $\eta = (1-\beta_1)\sum_{i\geq 0}\beta_1^i c_{-i}$ and $u_\alpha = \alpha\sum_{j\geq 0}\beta_2^j q_{-j}$. Group the double sum by $i-j$. If $i-j = s > 0$, the gradient comes first, the expectation is $C_s^{cq}(Q)$, and the total weight is $(1-\beta_1)\alpha\beta_1^s/(1-\beta_1\beta_2)$. If $j-i = s > 0$, the square comes first, the expectation is $C_s^{qc}(Q)$, and the weight is $(1-\beta_1)\alpha\beta_2^s/(1-\beta_1\beta_2)$. The diagonal gives the first term. For the expansion, $(1-\beta_1)\alpha/(1-\beta_1(1-\alpha)) = \alpha + O(\alpha^2)$ since $1-\beta_1 \geq 1-\bar{\beta}$, and $\sum_s |1-(1-\alpha)^s|\,|C_s^{qc}(Q)| \leq C\alpha$ by $1-(1-\alpha)^s \leq s\alpha$ and geometric decay. □

Lemma B.4 (Anchored cumulants). *Let $f_1, \ldots, f_r$, $2 \leq r \leq 4$, be bounded centered functions of a stationary finite irreducible, aperiodic chain. There is $C < \infty$ such that, for ordered times $t_1 \leq \cdots \leq t_r$,*

$$|\mathrm{cum}(f_1(X_{t_1}), \ldots, f_r(X_{t_r}))| \leq Ce^{-\max_j (t_{j+1}-t_j)/C}. \tag{B.13}$$

So with one time fixed, the cumulant is absolutely summable over the others, uniformly over the three kernels.

Proof. Bounds of this kind are standard for uniformly ergodic chains; we include the short argument for completeness. Write $Q^n = \Pi + R_n$ with $\|R_n\|_{\infty\to\infty} \leq Ce^{-n/C}$. Splitting the Markov product formula at a gap shows that a moment factorizes across that gap up to $Ce^{-\mathrm{gap}/C}$. For $r = 2, 3$ the cumulant is a moment, and every split leaves a centered singleton or a covariance across the gap. For $r = 4$, at the middle gap the moment minus the first pairing is $O(e^{-\mathrm{gap}/C})$ and the other two pairings each contain a covariance across that gap; at an outer gap the split product vanishes, because one side is a centered singleton, and every pairing contains a covariance across the gap. Taking the largest gap proves (B.13). With one time fixed and largest gap L, there are polynomially many tuples in $1+L$, which gives summability. □

Lemma B.5 (Nonlinear denominator). *For each kernel, uniformly in β_1,*

$$\mathbb{E}_{\nu_Q}[\eta\, d(\sigma^2 + u_\alpha)] = d'_*\mathbb{E}_{\nu_Q}[\eta u_\alpha] + O(\alpha^2), \qquad \bar{\Phi}'_{Q,\alpha}(0) = d_* + \lambda + O(\alpha). \tag{B.14}$$

Proof. Write $\eta = \sum_i e_i c_{-i}$ and $u_\alpha = \sum_j w_j q_{-j}$ with $e_i = (1-\beta_1)\beta_1^i$ and $w_j = \alpha(1-\alpha)^j \leq \alpha$, both summing to one. Keeping one weighted sum and bounding the other weight by α, the absolute covariance bounds give

$$\sum_{i,j} e_i w_j |\mathbb{E}[c_{-i}q_{-j}]| \leq C\alpha, \qquad \sum_{j,k} w_j w_k |\mathbb{E}[q_{-j}q_{-k}]| \leq C\alpha.$$

Lemma B.4, anchored at one time, makes every connected part with three or four factors carry an extra power of α, and the pairings are products of the two sums above. So

$\mathbb{E}[u_\alpha^2] = O(\alpha)$, $\mathbb{E}[u_\alpha^4] = O(\alpha^2)$, and $|\mathbb{E}[\eta u_\alpha^2]| + |\mathbb{E}[\eta u_\alpha^3]| = O(\alpha^2)$. Expanding $d(\sigma^2 + u_\alpha)$ to third order, with $\mathbb{E}[\eta] = 0$ and a bounded fourth derivative, gives the first claim. For the second, $\bar{\Phi}'_{Q,\alpha}(0) = \mathbb{E}[d(\zeta) + 2\eta\xi d'(\zeta)] + \lambda$ by (B.9), and $\mathbb{E}[\xi^2]$, $\mathbb{E}[\eta\xi]$, $\mathbb{E}[d(\zeta)] - d_*$, and $\mathbb{E}[\eta\xi(d'(\zeta) - d'_*)]$ are all $O(\alpha)$: the third because $\mathbb{E}[u_\alpha] = 0$, and the last by Cauchy–Schwarz, since $\mathbb{E}[|\xi u_\alpha|] \le (\mathbb{E}[\xi^2]\mathbb{E}[u_\alpha^2])^{1/2}$. □

Lemma B.6 (Stability for small α)**.** *There are $\alpha_0 > 0$ and $\omega_0 > 0$ such that, for $0 < \alpha \le \alpha_0$, each $\bar{\Phi}_{Q,\alpha}$ has a unique zero in $(-\bar{\theta}, \bar{\theta})$ and $\inf_{[-\bar{\theta},\bar{\theta}]} \bar{\Phi}'_{Q,\alpha} \ge \omega_0$, uniformly in β_1 and the kernel. In particular, Assumption 3.1 holds with $\omega_Q \ge \omega_0$.*

Proof. Since $\mathbb{E}[\xi^2] + \mathbb{E}[u_\alpha^2] = O(\alpha)$, comparing Φ and $\partial_\theta \Phi$ with their values at $(\xi, \zeta) = (0, \sigma^2)$, where both are affine in η and $\mathbb{E}[\eta] = 0$, gives

$$\bar{\Phi}_{Q,\alpha} \to \bar{\Phi}_0(\theta) = \theta d(\theta^2 + \sigma^2) + \lambda\theta \quad \text{in } C^1([-\bar{\theta}, \bar{\theta}]) \tag{B.15}$$

at rate $O(\sqrt{\alpha})$. The limit has $\bar{\Phi}'_0(\theta) = (\sigma^2 + \delta)(\theta^2 + \sigma^2 + \delta)^{-3/2} + \lambda \ge 2\omega_0 > 0$ on $[-\bar{\theta}, \bar{\theta}]$ and endpoint values of opposite signs, so for small α the derivative stays above ω_0 and the intermediate value theorem gives a unique zero. □

Proof of Theorem 3.4. By Lemma B.5 and (B.12), $\bar{\Phi}_{Q,\alpha}(0) = d'_* \mathbb{E}_{\nu_Q}[\eta u_\alpha] + O(\alpha^2) = O(\alpha)$ and $\bar{\Phi}'_{Q,\alpha}(0) = d_* + \lambda + O(\alpha)$. The slope bound of Lemma B.6 gives $|\theta_{Q,\alpha}| \le |\bar{\Phi}_{Q,\alpha}(0)|/\omega_0 = O(\alpha)$, and (B.10) is bounded uniformly, so a Taylor expansion at zero gives

$$\theta_{Q,\alpha} = -\frac{d'_*}{d_* + \lambda} \mathbb{E}_{\nu_Q}[\eta u_\alpha] + O(\alpha^2).$$

Substituting (B.12) proves (3.6). Lemma B.6 verifies Assumption 3.1, and Theorem 3.3 at fixed α gives (3.7); the range $\beta_1 \le \bar{\beta}$ keeps the mixing and tracking constants uniform in β_1. □

Proof of Corollary 3.5. Since $L(\theta) - L(0) = \theta^2/2$, squaring (3.6) gives (3.11). Use $\mathcal{T}_\Pi = \pi(c^3)$ for (3.12), and $\mathcal{T}_P = \mathcal{E}_{\beta_1} - \Gamma_{\beta_1}$, $\mathcal{T}_{P^\dagger} = \mathcal{E}_{\beta_1} + \Gamma_{\beta_1}$ for (3.13). With $\Delta = \theta - \theta_{Q,\alpha}$, Theorem 3.3 gives

$$\mathbb{E}[L(\theta)] - L(\theta_{Q,\alpha}) = \theta_{Q,\alpha}\mathbb{E}[\Delta] + \tfrac{1}{2}\mathbb{E}[\Delta^2] = O_\alpha(h).$$

This is uniform over the invariant laws in the class, so Lemma A.2 also gives the time-average bounds. Each nonzero leading coefficient determines the sign for small α, followed by small h. □

B.4 Proof of Proposition 3.2

Proof. Write $\underline{c} = \min c$ and $\overline{c} = \max c$, so that $\underline{c} < 0 < \overline{c}$ because c is centered and not constant, and let $J = [-\overline{c}, -\underline{c}]$. On a stationary path the filters η and ξ are weighted averages of past values of c, and ζ is the average of c^2 with the weights of ξ. So $\eta, \xi \in [\underline{c}, \overline{c}]$, and $\zeta - \xi^2$ is the variance of c under the weights of the variance memory. All statements about filter states below hold for ν_Q-almost every y.

Since $d'(v) = -\frac{1}{2}(v+\delta)^{-3/2}$ and $v_\theta + \delta - (\theta+\eta)(\theta+\xi) = (\zeta - \xi^2) + \delta + (\theta+\xi)(\xi-\eta)$, differentiating (3.3) gives

$$\partial_\theta \Phi(\theta, y) = \frac{(\zeta - \xi^2) + \delta + (\theta+\xi)(\xi-\eta)}{(v_\theta + \delta)^{3/2}} + \lambda. \tag{B.16}$$

Let $\theta \in J$. Then $\theta + \xi$ and $\xi - \eta$ both lie in $[\xi - \bar{c}, \xi - \underline{c}]$, an interval that contains zero, so their product is at least $-(\bar{c} - \xi)(\xi - \underline{c})$. If c only takes the values $\underline{c}$ and $\bar{c}$, a distribution on them with mean ξ has variance exactly $(\bar{c} - \xi)(\xi - \underline{c})$, and the numerator in (B.16) is at least δ. In general the variance is nonnegative and $(\bar{c} - \xi)(\xi - \underline{c}) \le \frac{1}{4}(\bar{c} - \underline{c})^2$, so the numerator is at least $\delta - \frac{1}{4}(\bar{c} - \underline{c})^2$. Set $\kappa = \delta$ when c takes two values and $\kappa = \delta - \frac{1}{4}(\bar{c} - \underline{c})^2$ otherwise. This lower bound is positive in both cases of the proposition. For $\theta \in J$ the variance v_θ is an average of $(\theta + c)^2 \le (\bar{c} - \underline{c})^2$, so

$$\partial_\theta \Phi(\theta, y) \ge \omega_1 := \frac{\kappa}{\{(\bar{c} - \underline{c})^2 + \delta\}^{3/2}} + \lambda, \qquad \theta \in J,$$

and the same bound holds for $\bar{\Phi}'_Q$.

Now let $-\underline{c} \le \theta \le 2\|c\|_\infty$. Then $\theta + \eta \ge 0$, $\theta > 0$, and $v_\theta \le 9\|c\|_\infty^2$, so $\Phi(\theta, y) \ge (\theta + \eta)\, d(9\|c\|_\infty^2)$. The filter η has mean zero under ν_Q, and therefore

$$\bar{\Phi}_Q(\theta) \ge \theta\, d(9\|c\|_\infty^2) \ge m_+ := |\underline{c}|\, d(9\|c\|_\infty^2) > 0.$$

In the same way $\bar{\Phi}_Q(\theta) \le -m_- := -\bar{c}\, d(9\|c\|_\infty^2)$ for $-2\|c\|_\infty \le \theta \le -\bar{c}$. So $\bar{\Phi}_Q$ is negative at the left end of J, positive at the right end, and strictly increasing in between, and its only zero θ_Q in $[-2\|c\|_\infty, 2\|c\|_\infty]$ lies in the interior of J. On J the mean value theorem gives $(\theta - \theta_Q)\bar{\Phi}_Q(\theta) \ge \omega_1(\theta - \theta_Q)^2$. Outside J, $\bar{\Phi}_Q(\theta)$ has the sign of $\theta - \theta_Q$ and absolute value at least $\min(m_+, m_-)$, while $|\theta - \theta_Q| \le 4\|c\|_\infty$. So (3.4) holds with

$$\omega_Q = \min\Big\{\omega_1,\ \frac{\min(m_+, m_-)}{4\|c\|_\infty}\Big\},$$

which only depends on c, δ, and λ. □

B.5 Proof of Proposition 3.6

Proof. Fix i and write ϑ for a value of $\partial_i L(\theta)$. By Assumption 3.1, $\bar{\Phi}_Q^{(i)}$ vanishes at $\theta_Q^{(i)}$ and at no other point of $[-2\|c_i\|_\infty, 2\|c_i\|_\infty]$. If $|\vartheta| > \|c_i\|_\infty$, then $\vartheta + \eta^i$ has the sign of ϑ on every path, and since $\lambda = 0$ so does $\bar{\Phi}_Q^{(i)}(\vartheta)$. Together with (3.4) this gives

$$\big(\vartheta - \theta_Q^{(i)}\big)\, \bar{\Phi}_Q^{(i)}(\vartheta) > 0 \qquad \text{for every } \vartheta \ne \theta_Q^{(i)}. \tag{B.17}$$

By (3.15) the zeros of $\bar{\Phi}_Q$ are then the solutions of $\nabla L(\theta) = r_Q$. For a strongly convex L the function $\theta \mapsto L(\theta) - \langle r_Q, \theta\rangle$ has exactly one critical point, its minimizer, so there is exactly one solution θ_Q.

Let μ be the modulus of strong convexity. Since $\nabla L(\theta_Q) = r_Q$, we have $D(\theta) \ge \frac{\mu}{2}|\theta - \theta_Q|^2$.

Also $\nabla D = \nabla L - r_Q$, so along the averaged flow

$$\frac{\mathrm{d}}{\mathrm{d}t} D(\theta) = -\sum_{i=1}^{n} \big(\partial_i L(\theta) - \theta_Q^{(i)}\big)\, \bar{\Phi}_Q^{(i)}\big(\partial_i L(\theta)\big).$$

By (B.17) every term of the sum is nonnegative, and the sum only vanishes when $\nabla L(\theta) = r_Q$, that is, at θ_Q. Where $|\partial_i L(\theta)| \le 2\|c_i\|_\infty$ for every i, (3.4) bounds the ith term below by $\omega_Q^{(i)}(\partial_i L(\theta) - \theta_Q^{(i)})^2$, which is (3.17). The sublevel sets of D are compact and forward invariant, the field $\bar{\Phi}_Q$ is continuously differentiable, and D decreases strictly away from θ_Q, so every solution of the flow converges to θ_Q. In the quadratic case $\nabla L(\theta) = A(\theta - \theta_*)$, which gives (3.18). □

C Example computations

C.1 The reversible two-state source

Explicitly, $P_\rho = \frac{1}{3}\left(\begin{smallmatrix} 2+\rho & 1-\rho \\ 2(1-\rho) & 1+2\rho \end{smallmatrix}\right)$, so its entries are positive exactly when $-1/2 < \rho < 1$. Since $\Pi f = \pi(f)\mathbf{1} = 0$ for centered f, we have $P_\rho f = \rho f$ and $P_\rho^s f = \rho^s f$. So $C_s^{cq}(P_\rho) = \pi(c\, P_\rho^s q) = \rho^s \pi(cq)$ and $C_s^{qc}(P_\rho) = \pi(q\, P_\rho^s c) = \rho^s \pi(cq)$, and with $\pi(cq) = \pi(c^3) = -2$ the geometric sums give (4.2). The matrix $\mathrm{diag}(\pi) P_\rho = \rho\, \mathrm{diag}(\pi) + (1-\rho)\pi^\top \pi$ is symmetric, so P_ρ is reversible. With $\delta = 1$ and $\sigma^2 = 2$, we have $d_* = 3^{-1/2}$ and $d_*'/d_* = -1/(2(\sigma^2 + \delta)) = -1/6$.

The multiplier in (4.3) equals $(1 - \beta_1\rho^2)/\{(1-\rho)(1-\beta_1\rho)\}$, which is positive, and its derivative in ρ is $(1-\rho)^{-2} + \beta_1(1-\beta_1\rho)^{-2} > 0$. Squaring the root ratio gives (4.4). At $\beta_1 = 1/2$ the multiplier is $31/45$ at $\rho = -1/4$ and $7/3$ at $\rho = 1/2$.

With $\lambda = 0$, Theorem 3.4 gives $\theta_{Q,\alpha} = \alpha \mathcal{T}_Q/6 + O(\alpha^2)$. So $\theta_{\Pi,\alpha} = -\alpha/3 + O(\alpha^2)$, and at $\rho = \beta_1 = 1/2$, $\mathcal{T}_{P_\rho} = -14/3$ and $\theta_{P_\rho,\alpha} = -7\alpha/9 + O(\alpha^2)$.

C.2 The symmetric cycle

Write the lazy cycle (4.5) as $P = (I + \mathsf{R})/2$, where $\mathsf{R}^3 = I$ and $\mathsf{R}^2 = \mathsf{R}^\top$. Its centered lag resolvent is

$$\sum_{s \ge 1} (P^s - \Pi) = \frac{\mathsf{R} - \mathsf{R}^\top}{3}. \tag{C.1}$$

It is skew-symmetric and π is uniform, so every centered vector observable Y satisfies $\Sigma_Y(P) = \Sigma_Y(P^\dagger) = \Sigma_Y(\Pi) = C_0^Y$. For $Y = (c, q)^\top$ from (4.5), put $C_s = \mathbb{E}_P[Y(X_0)Y(X_s)^\top]$. Geometric summation gives

$$C_0 = \begin{pmatrix} 2/3 & 0 \\ 0 & 2/9 \end{pmatrix}, \qquad \sum_{s \ge 1} C_s = \begin{pmatrix} 0 & 2/9 \\ -2/9 & 0 \end{pmatrix}, \tag{C.2}$$

which also identifies the opposite areas. The generating function is

$$G(t) = \sum_{s \ge 1} \gamma(s) t^s = \frac{2t}{3(t^2 - 2t + 4)}, \qquad 0 \le t \le 1, \tag{C.3}$$

which follows from the centered resolvent $t(P-\mathbf{1}\pi)[I-t(P-\mathbf{1}\pi)]^{-1}$ applied to c and q. Since $\Gamma_{\beta_1} = G(1) - G(\beta_1)$,

$$\frac{\Gamma_{\beta_1}}{\Gamma} = 1 - \frac{G(\beta_1)}{G(1)} = \frac{(1-\beta_1)(4-\beta_1)}{\beta_1^2 - 2\beta_1 + 4}.$$

With $\delta = 2/3$ we have $d'_*/d_* = -1/(2(\sigma^2+\delta)) = -3/8$, so $\alpha d'_* \Gamma/d_* = -\alpha/12$, and multiplying by Γ_{β_1}/Γ gives (4.6).

Lemma C.1 (Mirror symmetry)**.** *For the symmetric cycle, $\bar{\Phi}_{P^\dagger}(-\theta) = -\bar{\Phi}_P(\theta)$ for every θ, and the stable roots satisfy $\theta_{P^\dagger,\alpha} = -\theta_{P,\alpha}$. Also, $\pi(cq) = 0$ and $\chi(s) = 0$ for every s, so $\mathcal{E}_{\beta_1} = 0$.*

Proof. Let J swap states 0 and 2, so that $\mathsf{J}P\mathsf{J} = P^\dagger$ and $c \circ \mathsf{J} = -c$. Applying J to a whole source path gives a $P^\dagger$-path, and the causal filters change (η, ξ, ζ) into $(-\eta, -\xi, \zeta)$. So the image of ν_P is invariant for the reversed filters, and by uniqueness it equals $\nu_{P^\dagger}$. Since $v_{-\theta}(-\eta, -\xi, \zeta) = v_\theta(\eta, \xi, \zeta)$ and Φ changes sign, $\bar{\Phi}_{P^\dagger}(-\theta) = -\bar{\Phi}_P(\theta)$, and uniqueness of the stable zero gives $\theta_{P^\dagger,\alpha} = -\theta_{P,\alpha}$. The same swap keeps q and flips c, so $\mathbb{E}_P[c(X_0)q(X_s)] = -\mathbb{E}_{P^\dagger}[c(X_0)q(X_s)] = -\mathbb{E}_P[q(X_0)c(X_s)]$ by Lemma 2.1, which gives $\chi(s) = 0$. Finally $\pi(cq) = \pi(c^3) = 0$. □

C.3 An asymmetric cycle

On the symmetric cycle the two roots are mirror images, so reversal cannot change the loss. Changing the values of c breaks the symmetry. Keep the kernel in (4.5), $\delta = 2/3$, and $\lambda = 0$, and take

$$c = (4/3,\ 1/3,\ -5/3), \tag{C.4}$$

so that $\sigma^2 = 14/9$, $q = (2, -13, 11)/9$, and $\pi(cq) = \pi(c^3) = -20/27$. The covariance match persists, since it holds for every observable of this kernel. The centered resolvent gives

$$\sum_{s\geq 1} t^s \gamma(s) = -\frac{2t}{t^2 - 2t + 4}, \qquad \sum_{s\geq 1} t^s \chi(s) = \frac{20t(t-1)}{27(t^2-2t+4)}, \qquad 0 \leq t \leq 1, \tag{C.5}$$

and evaluating at $t = 1$ and $t = \beta_1$ yields

$$\mathcal{E}_{\beta_1} = -\frac{20}{27}\,\frac{4-\beta_1}{\beta_1^2 - 2\beta_1 + 4}, \qquad \Gamma_{\beta_1} = -\frac{2}{3}\,\frac{(1-\beta_1)(4-\beta_1)}{\beta_1^2 - 2\beta_1 + 4}. \tag{C.6}$$

With $d'_*/d_* = -1/(2(\sigma^2+\delta)) = -9/40$, and since $\mathcal{T}_P = \mathcal{E}_{\beta_1} - \Gamma_{\beta_1}$ and $\mathcal{T}_{P^\dagger} = \mathcal{E}_{\beta_1} + \Gamma_{\beta_1}$, Theorem 3.4 gives

$$\theta_{P,\alpha} = -\frac{(4-\beta_1)(1+9\beta_1)}{60(\beta_1^2 - 2\beta_1 + 4)}\,\alpha + O(\alpha^2), \qquad \theta_{P^\dagger,\alpha} = -\frac{(4-\beta_1)(19-9\beta_1)}{60(\beta_1^2 - 2\beta_1 + 4)}\,\alpha + O(\alpha^2). \tag{C.7}$$

Independent sampling has $\mathcal{T}_\Pi = \pi(cq) = -20/27$, so $\theta_{\Pi,\alpha} = -\alpha/6 + O(\alpha^2)$ for every β_1. To leading order, the forward root is closer to the minimizer exactly when $(4-\beta_1)(1+9\beta_1) < 10(\beta_1^2 - 2\beta_1 + 4)$, that is, $(1-\beta_1)(36-19\beta_1) > 0$, and the reversed root is farther out exactly when $(4-\beta_1)(19-9\beta_1) > 10(\beta_1^2 - 2\beta_1 + 4)$, that is, $(1-\beta_1)(36+\beta_1) > 0$. Both

hold for every $\beta_1 \in [0, 1)$. So to leading order all three roots sit below the minimizer, with the forward root closest and the reversed root farthest. For each fixed $\beta_1 < 1$, the forward source therefore gives a lower loss than independent sampling and the reversed source a higher one, once α and then h are small enough.

By Corollary 3.5, the two orientations differ by

$$L(\theta_{P,\alpha}) - L(\theta_{P^\dagger,\alpha}) = -\frac{(1-\beta_1)(4-\beta_1)^2}{20(\beta_1^2 - 2\beta_1 + 4)^2}\,\alpha^2 + O(\alpha^3). \tag{C.8}$$

The leading term is negative for every $\beta_1 \in [0, 1)$, so at each fixed β_1 the gap is negative once α is small enough. Since $L(\theta) - L(0) = \theta^2/2$, the excess losses have ratio

$$\frac{L(\theta_{P^\dagger,\alpha}) - L(0)}{L(\theta_{P,\alpha}) - L(0)} = \Big(\frac{19 - 9\beta_1}{1 + 9\beta_1}\Big)^2 + O(\alpha). \tag{C.9}$$

The limiting ratio is about 6.95 at $\beta_1 = 1/2$ and tends to one as $\beta_1 \uparrow 1$. At $\beta_1 = 0$ it equals 361, because the orientation component nearly cancels the shared part in the forward root, and the $O(\alpha)$ correction is then large.

D More numerical checks

The sticky source. Each comparison uses 3×10^4 coupled replicate pairs per step size, all started at $\theta = m = 0$ and $v = \sigma^2$, burn-ins of $\lceil 12/(hd_*)\rceil + \lceil 12/(1-\beta_2)\rceil$ steps, and windows of 2×10^4 steps. The runs use $\beta_2 = 0.99$ at six step sizes from $h = 0.02$ down to 0.000625, halving each time, with quadratic fits. The two smallest steps were added after the first four showed curvature.

Fitted separately, the i.i.d. means extrapolate to $(-3.35 \pm 0.04) \times 10^{-3}$ and $(-3.32 \pm 0.04) \times 10^{-3}$, against the linearized roots -3.33×10^{-3} and -3.30×10^{-3} for $\beta_1 = 0$ and $1/2$. The small momentum effect on the i.i.d. baseline is consistent with Theorem 3.4, where it enters at order α^2.

For the differences, three alternative fits (a quadratic without the largest step, a linear fit on the three smallest steps, and a cubic on all six) give intercepts between -3.26 and -3.24 for RMSprop and between -4.35 and -4.33 for Adam, in units of 10^{-3}. All three keep a nonzero intercept with the predicted sign, but their spread does not bound the extrapolation error.

The symmetric cycle. The runs use $\beta_2 = 0.99$, 8192 replicates at $h \in \{0.02, 0.01, 0.005, 0.0025, 0.00125\}$, 50-unit burn-ins, and a 200-unit window, where one unit is h^{-1} steps. The main fits are quadratic in h. Dropping the largest step or fitting a line to the three smallest changes the intercepts by about 0.2×10^{-5} or less, for either denominator. Figure 2 shows the main fits, and its bars measure Monte Carlo uncertainty only.

Frozen-root estimates. On the symmetric cycle, degree-16 moment closure with a truncated denominator expansion gives numerical estimates of the frozen roots. Comparing

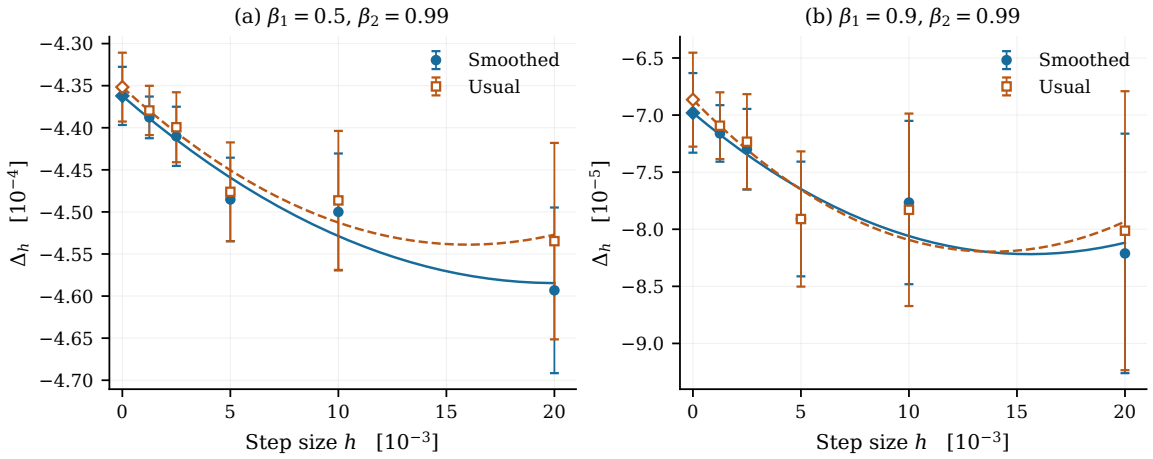


Figure 2: The symmetric cycle at $\beta_2 = 0.99$, with $\beta_1 = 1/2$ (left) and $9/10$ (right). The half-difference Δ_h of the forward and reversed means for both denominators, with fitted intercepts at $h = 0$. Bars show two Monte Carlo standard errors. The usual-denominator runs share source paths with the smooth runs, so their errors are correlated.

degrees and precisions and checking the residual of the truncated drift give diagnostics, but no certified error bound for the roots. Figure 3 compares the estimates with the linearized root and the leading small-α term.

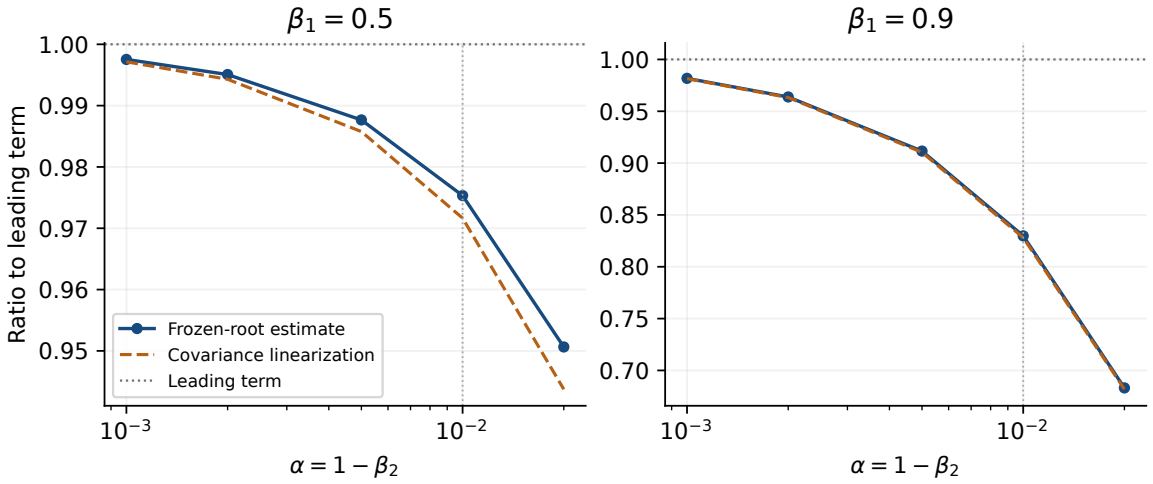


Figure 3: The fixed-memory orientation component on the symmetric cycle, divided by the leading small-α term. Solid curves are the moment-closure estimates of the frozen root, and dashed curves are the linearized root $-d'_*\mathbb{E}[\eta(\zeta - \sigma^2)]/d_*$. At $\alpha = 0.01$ (vertical dotted line), the estimates sit about 2.5% and 17% below the leading term for $\beta_1 = 1/2$ and 0.9. Panels use different vertical scales.

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